

Haitham's Sensors

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Haitham is a Computer Systems Engineering student working on an environmental monitoring system. A sensor continuously reports measurements, which are stored in chronological order. The first measurement establishes both the highest and lowest recorded values. For each subsequent measurement:

- If the measurement is **strictly greater** than the current highest value, a **new maximum** record is set.
- If the measurement is **strictly less** than the current lowest value, a **new minimum** record is set.

Given the sequence of sensor measurements, determine **how many times** the system records a **new minimum value** and how many times it records a **new maximum value**, respectively.

Input

The first line contains an integer n ($1 \leq n \leq 50$), the number of measurements. The second line contains n integers a_i ($-10 \leq a_i \leq 10$), where a_i is the i -th measurement.

Output

First print the number of times a new minimum record is set, then the same for a maximum, separated by a space

Example

standard input	standard output
6 3 1 2 6 4 3	1 1